

COURSE: CS 320 — Systems Programming

Credits: 3 | Department: Computer Science

Document ID: CAT-COURSE-CS320

Source: <https://catalog.techstate.edu/courses/cs320>

Date Accessed: March 29, 2025

PREREQUISITES: CS 210 AND CS 220

CO-REQUISITES: None

OFFERED: Fall, Spring

DESCRIPTION:

Low-level systems programming in C and C++. Topics include memory management (malloc, free, smart pointers), system calls (POSIX API), processes and inter-process communication (pipes, sockets, shared memory), multithreading (pthreads), signal handling, dynamic linking, debugging tools (GDB, Valgrind), and performance profiling.

GRADING: Programming projects 70%, Midterm 15%, Final 15%.